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MIL-A-85097/17(OS)
5 December 1994

MILITARY ASSOCIATED DETAIL SPECIFICATION

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTORS, UNDERSEAT, GAS INITIATED MK 123 MOD 0 AND MK 124 MOD 0

This specification is approved for use within the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Mk 123/124 Mod 0 Underseat Rocket Motors described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification covers requirements unique to two types of Under Seat Rocket Motors (USRMs).

1.2 Classification. The rocket motors shall be of the following types, as specified in the contract or purchase order (see 6.2.1).

Type I	Rocket Motor Mk 123 Mod 0
Type II	Rocket Motor Mk 124 Mod 0.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the basic specification, MIL-A-85097, and form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

MBEU142801	Rocket Motor Mk 123 Mod 0
MBEU142802	Rocket Motor Mk 124 Mod 0
RU-ATP-8	Acceptance Test Procedure for NACES Underseat Rocket Motors

(Application for copies of specifications and drawings should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Code 8420, 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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3. REQUIREMENTS

3.1 General. The requirements in this associated detail specification, unless in direct conflict with the basic specification, shall be considered additional, not substitute. USRMs manufactured to this specification shall comply with the requirements specified in Drawings MBEU142801 (Mk 123) and MBEU142802 (Mk124), Acceptance Test Procedure RU-ATP-9, associated process specifications, and table I. If there is a direct conflict with the basic specification, this document shall take precedence.

3.2 First article.

3.2.1 Propellant aging. When the loaded motor is tested per MIL-A-85097, it shall meet the ballistic parameters listed in Table I. The propellant shall meet the physical and mechanical properties of the manufacturer's propellant specification for a minimum of 84 months from the month of extrusion.

3.3 First article/production.

3.3.1 Propellant. The main propellant charge used in the USRM shall be KU (produced by Royal Ordnance, plc, Bishopton, Scotland), or a Navy-approved equivalent. The propellant shall conform to RO's internal propellant specification (ref: BS-T-480-S05-20129-1) or a Navy-approved equivalent when
tested for MIL-A-85097 data. The propellant manufacturer shall make the results of the physical and mechanical tests available to the rocket motor manufacturer(s) for comparison (see 6.2.2).

3.3.2 Life. Life shall conform to the requirement of MIL-A-85097, with the exception of propellant life which shall be 3 years maximum on sheet stock and 1 year maximum on extruded propellant grains.

3.3.3 Ballistic Performance. Rocket Motors Mk 123 Mod 0 and Mk 124 Mod 0 shall meet the ballistic performance requirements specified in Table I when tested in accordance with 4.3.3 .

TABLE I. Ballistic performance requirements for Mk 123/124 Mod 0 Rocket Motors (types I and II).

Parameter	165 ± 5°F		-65 ± 5°F	
	Min	Max	Min	Max
Ignition delay, sec	-	.010	-	.010
Lateral thrust operation time, sec	.049	.074	.060	.101
Initial impulse, lb-sec	235	381	235	381
Initial thrust, lb	4593	5408	3303	4179
Total impulse, lb-sec	1116	1233	1116	1233
Stable thrust, lb	4243	5456	3532	4353
Action time, sec	.226	.263	.269	.321

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Maximum thrust, lb	4939	6582	3837	4622
Initiation pressure, psig	400	600	400	600

4. QUALITY ASSURANCE PROVISIONS

4.1 Production Quality Conformance Inspection.

4.1.1 Sampling. For sampling purposes, Mk 123 and Mk 124 rocket motor lots may be considered a single lot if they are loaded concurrently with the same propellant and cartridge assembly lots. Individual sample quantities of the various rocket motors comprising the total lot acceptance sample shall be proportionate to the corresponding quantity in the delivered lot.

4.2 First article vibration testing. For the AEPS covered by this specification sheet, the following test procedures shall be used in lieu of those specified in the general specification.

a. Prior to vibration testing, the rocket motors shall be temperature conditioned for a minimum of 6 hours and a maximum of 48 hours, one at -65 ± 5°F and the other at 165 ± 5°F. The rocket motors shall be maintained at these conditioning temperatures during vibration tests. A continuous temperature history of the rocket motor shall be kept for all conditioning and test times.

b. If the rocket motor is removed from the conditioning chamber, or if the rocket motor temperature falls outside the specified temperature limits for more than 5 minutes, the testing shall stop and the rocket motor shall be reconditioned. A reconditioning time of 2 minutes is required for every minute the rocket motor spent outside the specified temperature envelope. Reconditioning time shall not exceed 6 hours.

c. Excitation shall be applied through the aircraft seat attachment fittings unless otherwise noted in the contract (see 6.2.1). Accelerometers shall be placed midway between the attachment points to measure acceleration along the axis perpendicular to the axis through the attachment points of the rocket motor. Longitudinal vibration shall be measured at either end of the rocket motor.

d. Each test unit shall undergo random vibration testing per the "Aircraft with jet engines" vibration test of MIL-STD-2102. A successful candidate item shall meet the vibration criteria specified in MIL-STD-2102.

4.3 First article/production.

4.3.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: -65 ± 5°F
b. Hot: 165 ± 5°F

4.3.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

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a. Static firing: 6 to 48 hours
b. Temperature cycling: 6 to 8 hours

4.3.3 USRM test (types I & II). The USRM shall be temperature conditioned a minimum of six hours and not more than 48 hours prior to testing. Within five minutes of removal from the conditioning chamber, the unit shall undergo ballistic testing on a three component test stand. The following ballistic data shall be recorded: ignition delay, lateral thrust motor operation time, initial impulse, initial thrust, total impulse, stable thrust, action time, maximum thrust, initiation pressure, and thrust-time curve.

5. PACKAGING

(USRMs shall be packaged per MIL-A-85097 and the applicable contract.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rocket motors described herein are intended for use with SJU-17(V) Navy Aircrew Common Ejection Seat (NACES) as a means of supplementing upward thrust during the ejection sequence. They are used in the T-45A, F/A-18C/D, and F-14D aircraft as indicated in Table II.

TABLE II. USRM application.

Seat Number	Aircraft	Location	Motor Type
SJU-17(V)1/A	F/A-18C*	-	II
SJU-17(V)1/A	F/A-18D	Rear	II
SJU-17(V)2/A	F/A-18D	Front	I
SJU-17(V)3/A	F-14D	Front	I
SJU-17(V)4/A	F-14D	Rear	II
SJU-17(V)5/A	T-45	Front	I
SJU-17(V)6/A	T-45	Rear	II

*Single seat aircraft

6.1.1 General description.

6.1.1.1 Type I: Mk 123 Mod 0 (USRM). The USRM is a ballistic gas actuated solid propellant rocket motor. The USRM contains an integral divergence rocket motor. The divergence motor on the MK 123 causes the pilot to diverge to the left when looking from the cockpit to the nose of the aircraft.

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The unit generates sufficient thrust and divergence to propel the man/seat mass to an altitude sufficient for safe parachute descent.

6.1.1.2 Type II: Mk 124 Mod 0 (USRM). The Mk 124 is almost the same as the Mk 123 described in 6.1.1.1. The only major difference is the side of the rocket motor on which the integral divergence rocket motor is located. The Mk 124 causes the pilot to diverge to the right when looking from the cockpit to the nose of the aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Mk and Mod designation of rocket motor.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52-209-3 and -4 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirements title	Applicable DID no.
3.3.1	Acceptance Test Report	UDI-T-22741A

(Data items descriptions related to this specification, and identified in this section, will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions. The following ballistic performance definitions apply to the USRM.

6.3.1 Ignition delay (sec). The elapsed time from ignition signal (shear pin shear) to first indication of continuous thrust on the Fz thrust-time curve.

6.3.2 Lateral thrust motor burn-out point. The intersection of the tangent to the declining portion of the lateral thrust (Fy) curve with the Fy zero thrust line.

6.3.3 Lateral thrust motor operation time (sec). The time measured between the first rise of the Fz curve and the lateral thrust motor burn-out point.

6.3.4 Initial impulse (lb-sec). The area under the Fz thrust curve measured between the initial rise and the lateral thrust motor burn-out point, multiplied by its horizontal and vertical scales.

6.3.5 Initial thrust (lb). Initial impulse divided by lateral thrust motor operation time.

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6.3.6 Total impulse (lb-sec). The total area under the Fz thrust curve multiplied by its horizontal and vertical scales.

6.3.7 Stable thrust period. The period between 0.150 and 0.200 seconds measured from first rise of Fz thrust.

6.3.8 Stable thrust (lb). The mean thrust over the stable thrust period.

6.3.9 Action time (sec). The time interval from 50 percent of stable thrust on the initial rise of the Fz thrust-time curve to the corresponding 50 percent of stable thrust on the declining portion of the curve.

6.3.10 Maximum thrust (lb). The maximum thrust measured.

Preparing Activity:
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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-85097/17	2. DOCUMENT DATE (YYMMDD) 5 December 1994

3. DOCUMENT TITLE
AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTORS, UNDERSEAT, GAS INITIATED MK 123 MOD 0 AND MK 124 MOD 0

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-4700 (2) DSN 354-4700
c. ADDRESS (*Include Zip Code*) Standardization Team (Code 840M) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office, Attn: DLSC-LM 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

DD Form 1426, OCT 89 *Previous editions are obsolete.*

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